

Intrabeam Collisions, Gas and Electron Effects

USPAS - Beam Physics with Intense Space Charge

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Coulomb collisions

Gas and electron effects

- ▶ Effects depend strongly on particle mass and charge.
- ▶ Timescales are much different in circular accelerators vs. linacs ($t_{residence} \sim$ milliseconds to days vs. 10's of microseconds).
- ▶ Long vs. short pulse ($t_{pulse} \sim$ 10's of microseconds vs. 10's of nanoseconds).

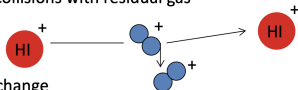
 Heavy ion	 Residual gas molecule	e^- electron
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Processes:

1. Coulomb collisions (intra-beam)



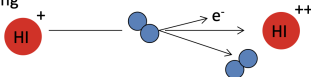
2. Coulomb collisions with residual gas



3. Charge exchange



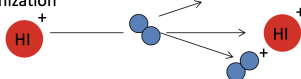
4. Stripping



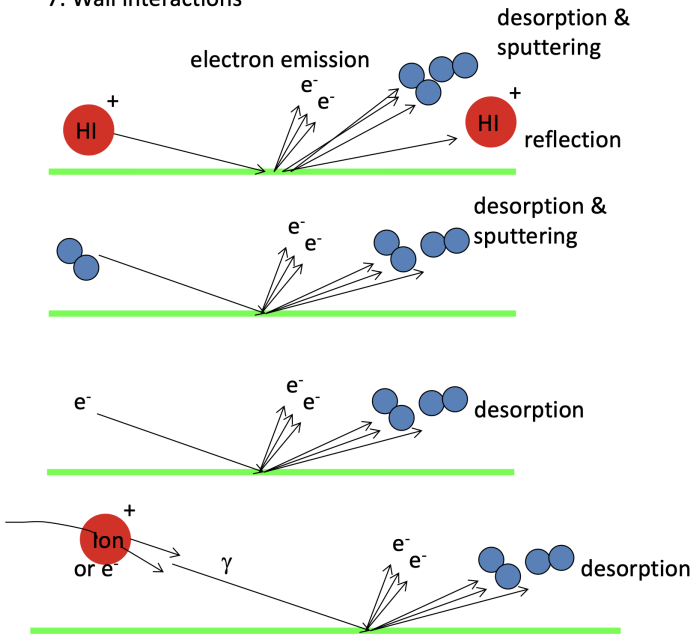
5. Neutralization



6. Gas Ionization



7. Wall interactions



Coulomb collisions

Consider effects of Coulomb collisions in a continuous beam propagating through a continuous focusing channel with $T_{\perp} \neq T_{\parallel}$. (If $T_{\perp} \neq T_{\parallel}$, the beam is already relaxed.)

From Ichimaru and Rosenbluth [1]:

$$\frac{dT_{\perp}}{dt} = -\frac{1}{2} \frac{dT_{\parallel}}{dt} = \frac{T_{\parallel} - T_{\perp}}{\tau}.$$

(Since $T_x = T_y = T_{\perp}$, $T_{\parallel}$ changes at twice the rate of $T_{\perp}$. And since $2k_B T_{\perp} + k_B T_{\parallel} = \text{constant}$.)

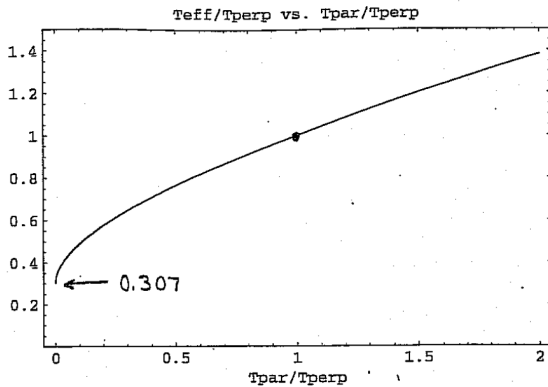
τ = relaxation time

$$\begin{aligned} &= \frac{15(k_B \tilde{T}/mc^2)^{3/2} (4\pi\epsilon_0)^2 m^2 c^3}{8\pi^{1/2} q^4 \ln \Lambda n} \\ &= \left(\frac{15\pi^{1/2}}{8 \ln \Lambda} \right) \frac{1}{\nu_c}, \end{aligned}$$

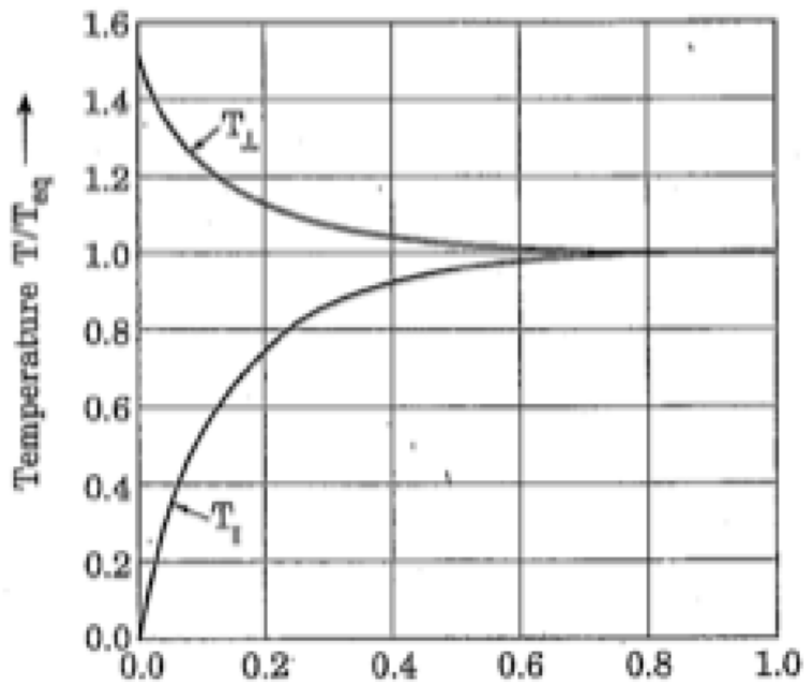
where ν_c is the collision frequency at thermal equilibrium:

$$\nu_c \sim \pi \left(\frac{q^2}{4\pi\epsilon_0 k_B T} \right)^2 n_0 \left(\frac{k_B T}{m} \right)^{1/2}.$$

$$\frac{T_{\text{eff}}}{T_{\perp}}$$



[...]



The Boersch effect

Aren't collisions negligible? Putting in numbers for ions:

$$\tau_{\text{eff}} = 4.3 \cdot 10^{-4} [s] \left(\frac{A^{1/2}}{Z^4} \right) \left(\frac{k\tilde{T}}{[eV]} \right)^{3/2} \left(\frac{15}{\ln \Lambda} \right) \left(\frac{10^{10} [cm^{-3}]}{n} \right) \quad (1)$$

$$\Lambda = \frac{1.5 \cdot 10^5 (k_B T / [eV])^{3/2}}{Z^3 (n / 10^{10} [cm^{-3}])} \quad (2)$$

Example: 2 MeV injector:

[...]

Beam-gas scattering

Coulomb collisions in residual gas

Rate of change of momentum transverse to velocity:

$$\frac{dp_x}{dt} = \frac{ZZ_g e^2}{4\pi\epsilon_0 r^2} \frac{b}{r}$$

Total change in momentum:

$$\begin{aligned}\Delta p &= \int_{-\infty}^{\infty} \frac{dp_x}{dt} dt \\ &= \int_{-\infty}^{\infty} \frac{dp_x}{dt} \frac{dt}{dz} dz \\ &= \frac{ZZ_g e^2 b}{4\pi\epsilon_0 v} \int_{-\infty}^{\infty} \frac{dz}{(z^2 + b^2)^{3/2}} \\ &= \frac{2ZZ_g e^2 b}{4\pi\epsilon_0 v b}.\end{aligned}$$

Approximate deflection angle θ :

$$\theta \approx \frac{\Delta p}{p} = \frac{2ZZ_g e^2 b}{4\pi\epsilon_0 p v b}$$

Multiple collisions

After traversing distance s and undergoing N_s collisions, the mean square angle $\langle \Theta^2 \rangle$ can be written

$$\begin{aligned}\langle \Theta^2 \rangle &= N_s \langle \theta \rangle^2 \\ &= n_g \sigma_s s \langle \theta \rangle^2 \\ &= 8\pi n_g \left(\frac{ZZ_g e^2}{4\pi\epsilon_0 mc^2 \gamma \beta^2} \right)^2 \ln \left(\frac{\theta_{max}}{\theta_{min}} \right) s\end{aligned}$$

Writing

$$\langle \Theta^2 \rangle = \langle x'^2 \rangle + \langle y'^2 \rangle = 2\langle x'^2 \rangle,$$

and taking the s -derivative gives a dynamical equation for $\langle x'^2 \rangle$:

$$\frac{d}{ds} \langle x'^2 \rangle = 4\pi n_g \left(\frac{ZZ_g e^2}{4\pi\epsilon_0 mc^2 \gamma \beta^2} \right)^2 \ln \left(\frac{\theta_{max}}{\theta_{min}} \right) \quad (3)$$

$$\equiv C. \quad (4)$$

Jackson's approximation of θ_{min} and θ_{max}

Jackson argues that θ_{max} arises from the distributed nature of the nucleus and θ_{min} from the screening of electrons or the quantum uncertainty principle.

$$\ln \left(\frac{\theta_{max}}{\theta_{min}} \right) \approx 2 \ln (204 Z_g^{-1/3})$$

How does scattering affect the envelope equations?

Thin lens approximation: assume the scattering locally changes the transverse momentum without directly changing the position.

Let the initial phase space coordinates be $x = x_0$ and $x' = x'_0$. After an incremental distance δs ,

$$x \rightarrow x_0 \tag{5}$$

$$x' \rightarrow x'_0 + \delta x'. \tag{6}$$

We can then calculate the incremental change in the second-order moments:

$$\text{delta}\langle x'^2 \rangle = \langle \dots \rangle$$

Charge-changing processes

Gas pressure instability

Electron cloud processes

Electron-ion instability

References

- [1] S. Ichimaru and M. N. Rosenbluth, *Relaxation Processes in Plasmas with Magnetic Field. Temperature Relaxations*, The Physics of Fluids **13**, 2778 (1970).